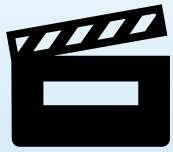


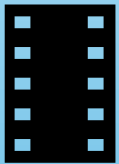
**PyData Boston 2025**

**Fun With Python and Em😊ji:  
What Might Adding Pictures  
to Text Programming  
Languages Look Like?**



## TODAY'S SCRIPT

- 1 THE BIG IDEA
- 2 A PICTURE IS WORTH 1,000 WORDS, RIGHT?
- 3 SO, WHAT MIGHT "EMOJICODE" LOOK LIKE?
- 4 CUTE, BUT IS IT USEFUL?
- 5 A RUDIMENTARY EMOJI-TO-TEXT PREPROCESSOR
- 6 THAT'S ALL FOLKS! (CONCLUSION, Q&A)





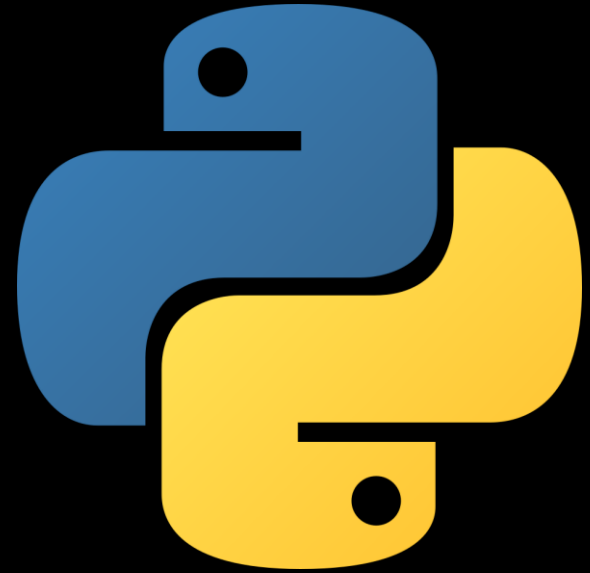
You see things; and  
you say "Why?" But I  
dream things that  
never were; and I say  
"Why not?"



**George Bernard Shaw**  
Irish Playwright

1856 - 1950

QUOTEHD.COM



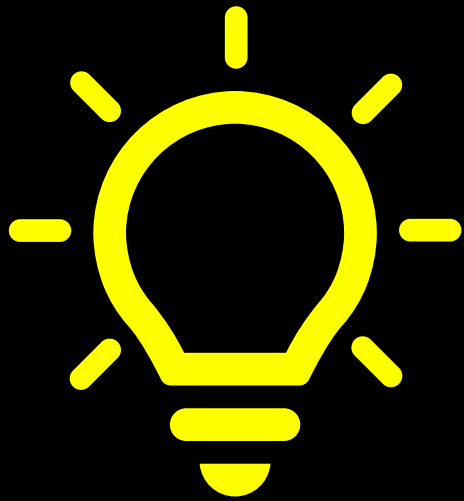
+



---

Emojis in code? Where did that  
Aha! moment come from?

---





## COPILOT-GENERATED SAS CODE

```
proc print data=sashelp.cars(obs=5);  
run;
```

```
proc freq data=sashelp.cars;  
tables type;  
run;
```

```
proc means data=sashelp.cars;  
var mpg_city;  
run;
```

```
proc univariate data=sashelp.cars;  
var horsepower;  
run;
```

```
proc sgplot data=sashelp.cars;  
scatter x=weight y=mpg_city;  
run;
```



**PROBLEM:** How can I stop all those pesky “**RUN;**” statements from cluttering up SAS code but leave the functionality intact for those who wish to use it?



**SOLUTION:** Replace the “**RUN;**” statements (and the “**;**” from the preceding statement) with an emoji, icon, image, or other pictogram that gets expanded to “**;; RUN;**”



= “**;; RUN;**”

ABC + 😊!

# DEBATE CLUB

ABC!

If you enjoy communicating and would like to improve your critical thinking skills **join debate club!**

Here we will learn how to debate,  
be persuasive and logically  
structure your thoughts.

When? Every Wednesday at 15.30  
Where? Room 4A





# ≡ Would You Use a "Running Semicolon" Character?

📅 Posted 02-20-2022 04:06 PM (3248 views)



*"Running Semicolon"*



As HBO's Silicon Valley amusingly showed us, programmers can get into heated arguments over the use of [Tabs vs. Spaces](#).

In the SAS world, debates can arise over [whether every DATA step and PROC should be terminated with a RUN or QUIT statement](#) - large SAS programs may include 100+ "RUN;" statements - many technically 'unnecessary' (depending on who you ask!) - making some SAS programmers happy while irritating others. So, a question for lovers and haters of the RUN-even-if-unnecessary rule: If a "running semicolon" or "RUN character" was available, would you be okay with the use of that? 👍👎

Join  
Ha

Cal  
sou  
pro  
the  
31st

How

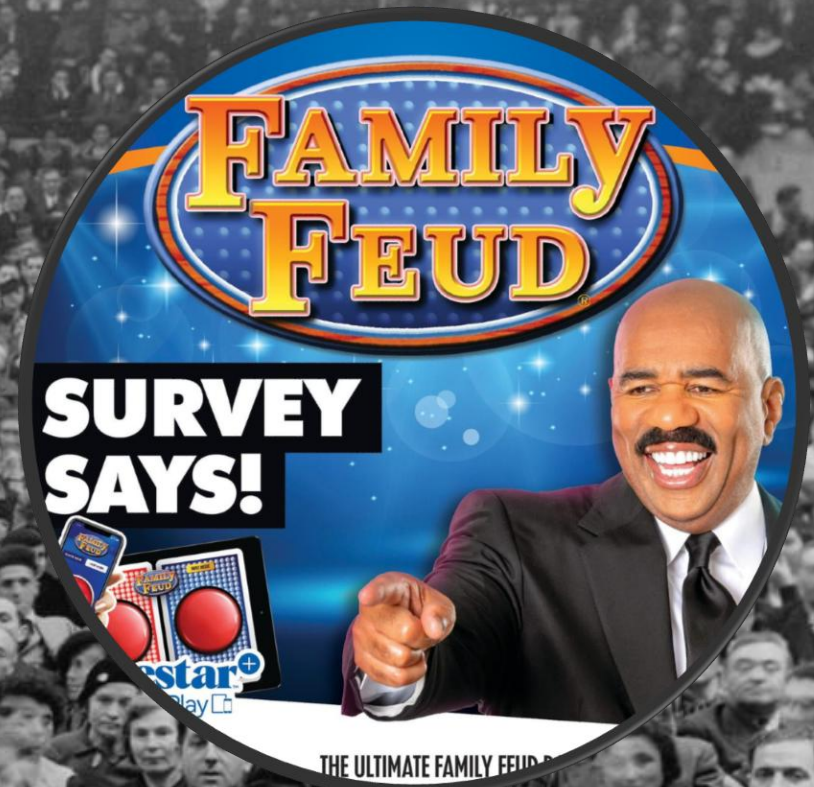
Learn  
SAS t  
variab

CO  
IN S

“I cringe at the notion...”

“Bad idea IMO...”

“Programming languages shouldn’t have decoration: save that for places like Twitter...”





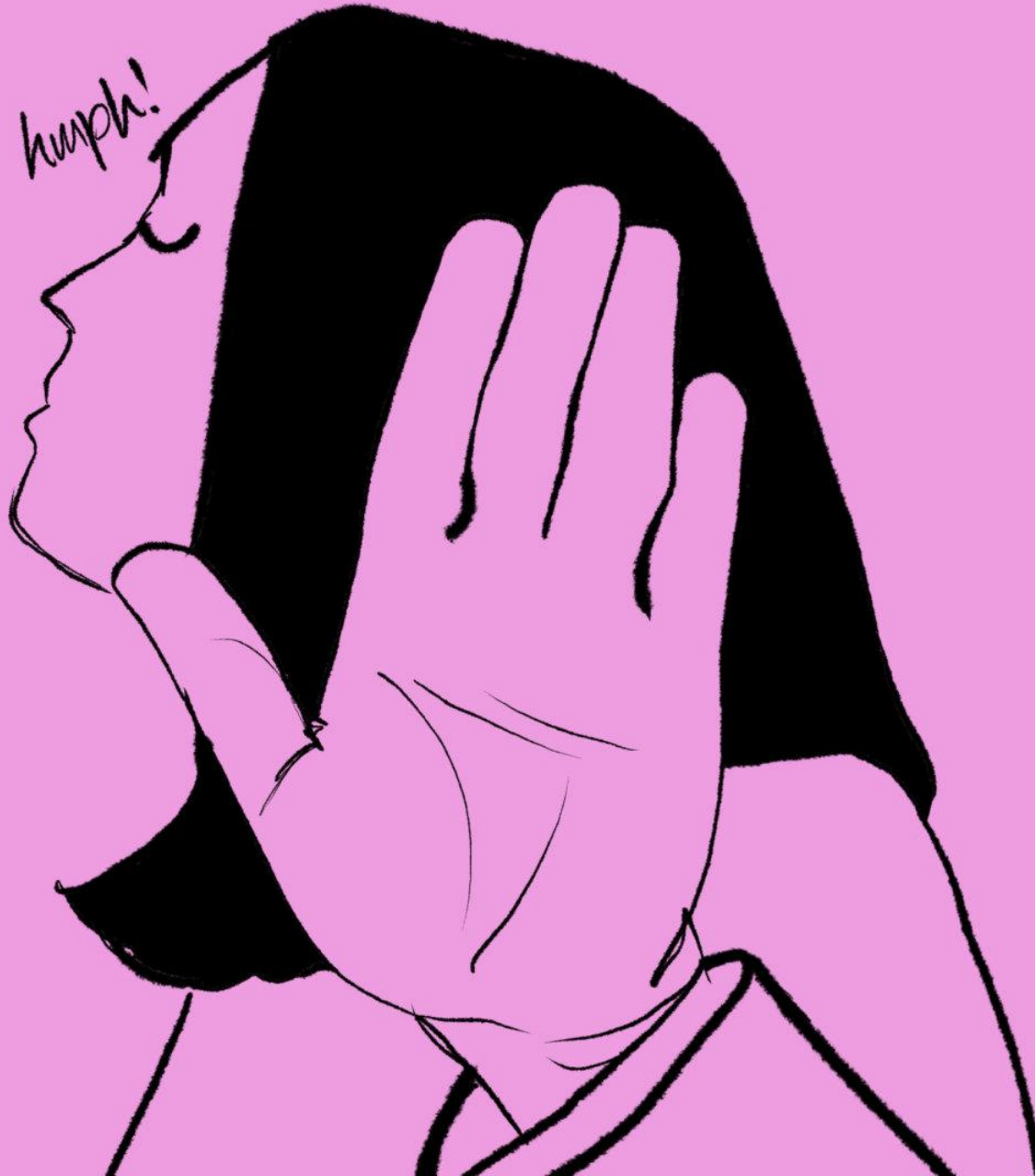
WHY THE PUSHBACK?

GANESH  
SUBRAMANIAN



SYMBOLS  
A PICTURE IS WORTH A THOUSAND WORDS

*Chaitanya*





IT'S A  
TIME-TESTED  
CONCEPT





**Aa**  
apple



**Bb**  
bear

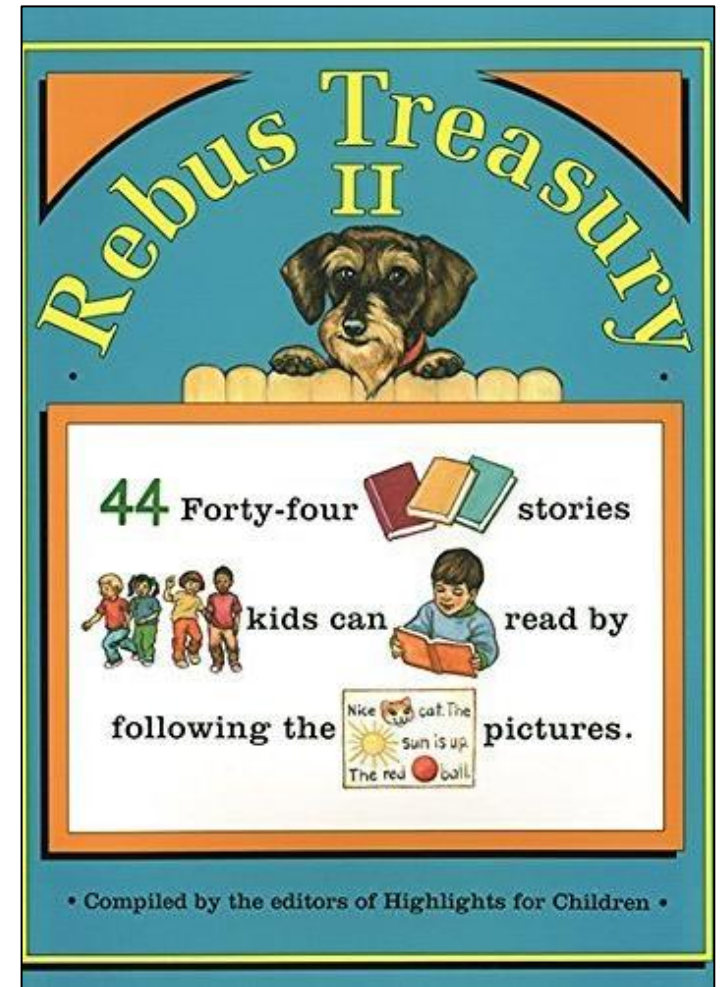
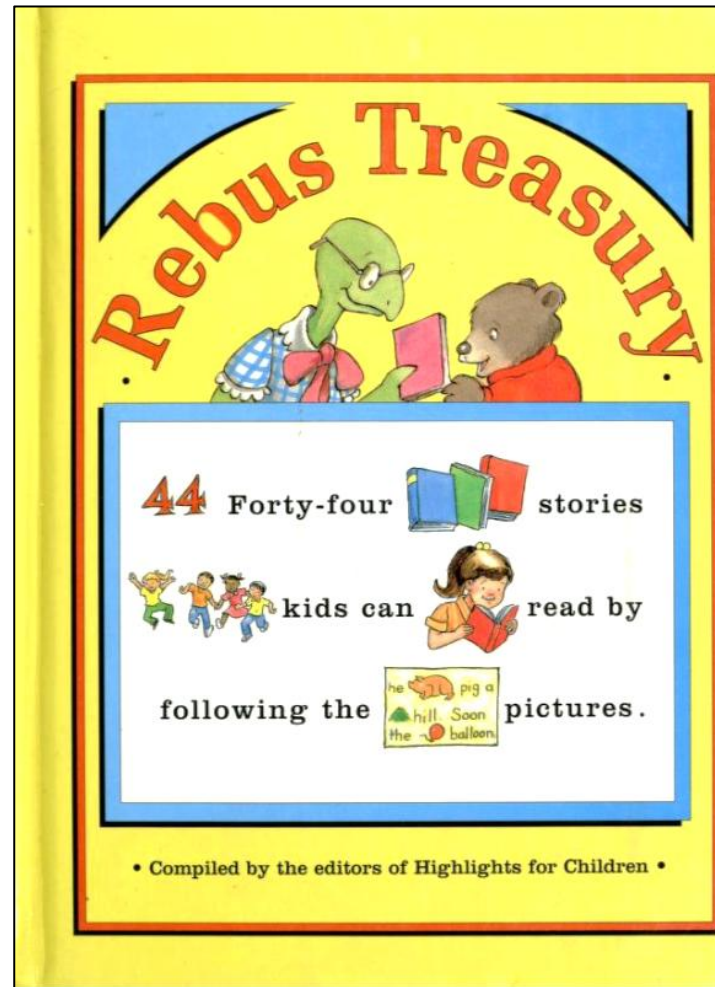


**Children learn to  
associate words  
and pictures at an  
early age**

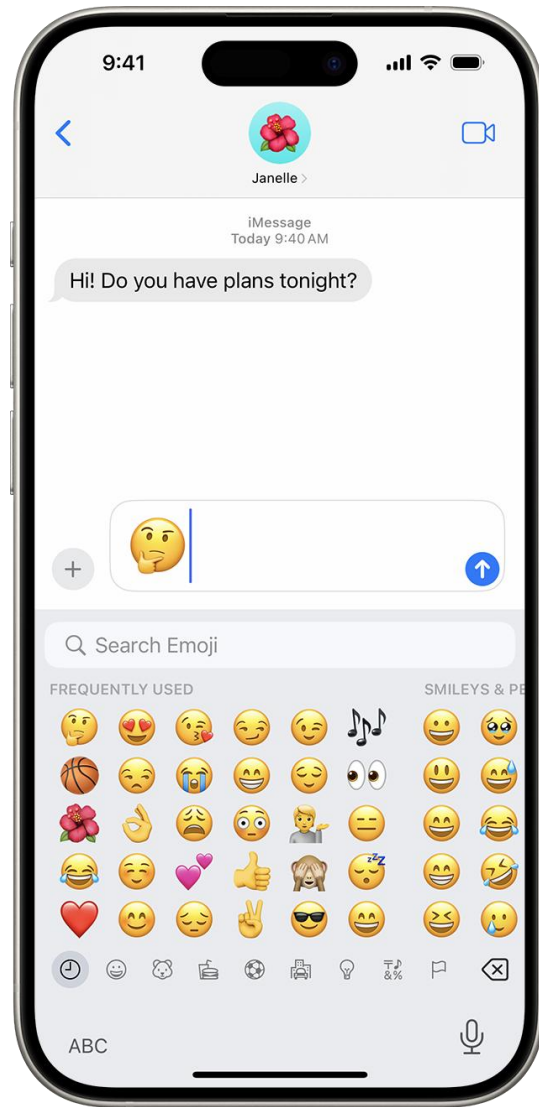


# Highlights for Children: Rebus Stories

- Use pictures or symbols to represent words or parts of words
- Designed for children just beginning to read
- Playful mix of text and images keeps reading fun, reduces frustration
- Visually engaging reading experiences

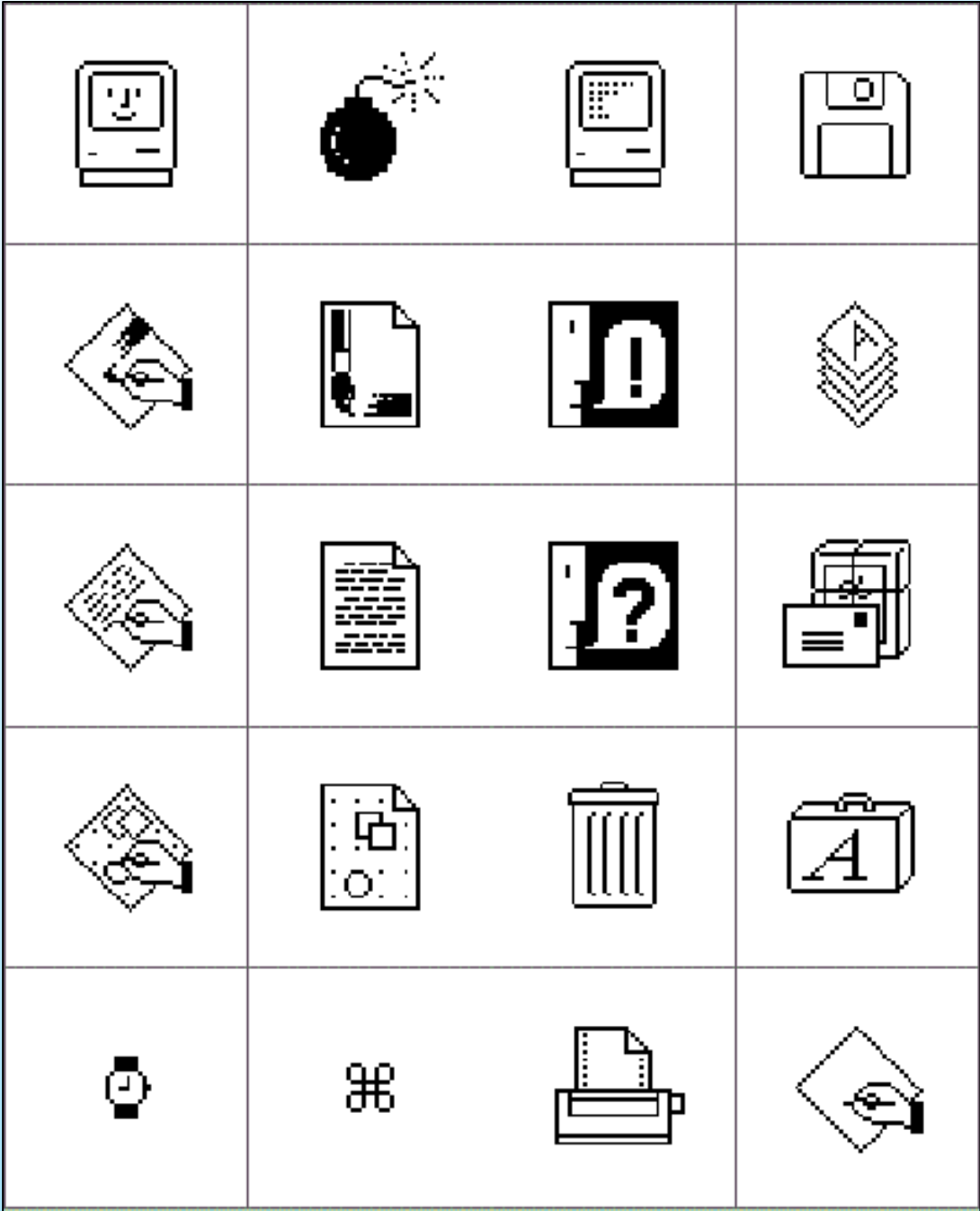






I ❤️ NY®

Baa, baa .  
Have you any  ?  
Yes, sir. Yes, sir.  
   full.



Characters Generated by  
Cairo Type Font

|           |   |   |   |   |   |   |   |   |   |   |   |        |            |
|-----------|---|---|---|---|---|---|---|---|---|---|---|--------|------------|
| ~         | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 0 | - | =      | Back Space |
| 🍌         | 🌴 | 👁 | 🪑 | 🍷 | 🍰 | 🍷 | 🌙 | ⭐ | 🌊 | ✂ | ➡ | ⬅      |            |
| Tab       | a | w | e | r | t | y | u | i | o | p | f | j      | \          |
|           | ✈ | ☘ | 🐢 | ⚡ | ⚓ | 📱 | 💣 | 🍷 | 🌍 | 👉 | ⬇ | 🐲      | 🖨          |
| Caps Lock | a | s | d | f | g | h | j | k | l | ; | ' | Return |            |
|           | ✍ | ⚡ | ✳ | 🔑 | 🌀 | ➡ | 📱 | 🔑 | 🚗 | 📱 | 📱 |        |            |
| Shift     | z | x | c | v | b | n | m | , | . | / | + | Shift  |            |
|           | 🐶 | ✚ | 🦋 | 💡 | 👓 | 👤 | 👤 | 📱 | 📱 | 📱 | 📱 |        |            |
| Opt       | 🍌 |   |   |   |   |   |   |   |   |   |   | Enter  | Opt        |

|                                                                                     |                                                                                       |                                                                                       |                                                                                       |                                                                                       |                                                                                       |                                                                                       |                                                                                       |                                                                                       |                                                                                       |                                                                                       |                                                                                      |                                                                                      |                                                                                     |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| ~                                                                                   | !                                                                                     | @                                                                                     | #                                                                                     | \$                                                                                    | %                                                                                     | &                                                                                     | *                                                                                     | (                                                                                     | )                                                                                     | -                                                                                     | +                                                                                    | Back Space                                                                           |                                                                                     |
|  |    |    |    |    |    |    |    |    |    |    |   |                                                                                      |                                                                                     |
| Tab                                                                                 | Q                                                                                     | W                                                                                     | E                                                                                     | R                                                                                     | T                                                                                     | Y                                                                                     | U                                                                                     | I                                                                                     | O                                                                                     | P                                                                                     | {                                                                                    | }                                                                                    |                                                                                     |
|                                                                                     |    |    |    |    |    |    |    |    |    |    |   |   |  |
| Caps Lock                                                                           | A                                                                                     | S                                                                                     | D                                                                                     | F                                                                                     | G                                                                                     | H                                                                                     | I                                                                                     | K                                                                                     | L                                                                                     | ;                                                                                     | "                                                                                    |                                                                                      | Return                                                                              |
| Lock                                                                                |   |   |   |   |   |   |   |   |   |   |  |  |                                                                                     |
| Shift                                                                               | Z                                                                                     | X                                                                                     | C                                                                                     | V                                                                                     | B                                                                                     | N                                                                                     | M                                                                                     | <                                                                                     | >                                                                                     | ?                                                                                     |                                                                                      |                                                                                      | Shift                                                                               |
|                                                                                     |  |  |  |  |  |  |  |  |  |  |                                                                                      |                                                                                      |                                                                                     |
| Opt                                                                                 |  |                                                                                       |                                                                                       |                                                                                       |                                                                                       |                                                                                       |                                                                                       |                                                                                       |                                                                                       |                                                                                       |                                                                                      | Enter                                                                                | Opt                                                                                 |

To Insert These Characters in Your Text  
Select "Cairo" Font, Size 18 Point





life $\leftarrow\{\uparrow 1 \omega V.\wedge 3 4=+/,^{-} 1 0$   
 $1\circ.\ominus^{-} 1 0 1\circ.\phi\subset\omega\}$

APL – A PROGRAMMING  
 LANGUAGE (Early 1960s)

**THREE  
YEARS  
LATER...**



Paper 184-2025

# Fun With SAS® and Emoji

What Might a Rebus-Influenced Programming  
Language Look Like?



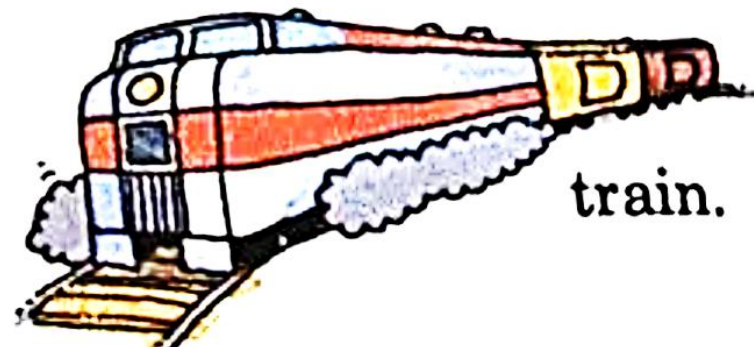
Highlight

# Jimmy Goes to the Library

By Pat Kite



Jimmy wanted a book about a



train.



Mother said.

They took a



bus to the library.

and the book I want?"



Jimmy asked.



if g=h then i=j; else k=substr(L, 1, 3);



g=h



i=j;



k=



(L, 1, 3);

**IF-THEN-ELSE**



## SAS Code (Text)

```
data test;  
set;  
if a=b then c=d; else e=f;  
if g=h then i=j; else k=substr(L, 1, 3);  
dt=date(); tm=time();  
proc print; run;
```

## SAS Code (Text+Emoji)

```
💾 test;  
set;  
🤔 a=b ✅ c=d; ❌ e=f;  
❓ g=h 👍 i=j; 👎 k=✂️ (L, 1, 3);  
dt=📅 (); tm=🕒 ();  
proc 🖨️ ; 🏃
```

# SQL

**select make from cars group by 1 order by 1 desc;**



**make**



**cars**

**$\Sigma$**

**1**



**1**



**;**

**SELECT**

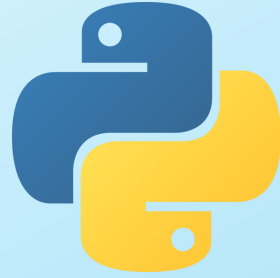


## SQL Code (Text)

```
proc sql noprint;
create table car_summary as
select make, model,
max(dt) as maxdate format=date10.,
min(tm) as mintime format=time8.
from sashelp.cars t1
join dt_tm t2 on 1=1
group by make, model
order by make, model desc;
```

## SQL Code (Text+Emoji)

```
proc sql noprint;
🗄 car_summary as
🛒 make, model,
max(dt) as max 📅 format=📅 10.,
min(tm) as min 🕒 format=🕒 8.
👈 sashelp.cars t1
🔗 dt_tm t2 on 1=1
Σ make, model
📶 make, model 📶 ;
```



```
if datetime.now().weekday() in (5, 6): print ("Weekend!")
```



```
.now().weekday() in (5, 6): 🖨️ ("Weekend!")
```



## IF-THEN-ELSE

## Python Code (Text)

```
import pandas, duckdb
Import plotly.express as plotly
from datetime import datetime
print(datetime.now().strftime("%Y-%m-%d"))
if datetime.now().weekday() in (5, 6):
    print ("It's the weekend!")
else:
    print("It's a work day!")

df_cars=pandas.read_csv('cars.csv')
plotly.bar(cars_summary, x='Type',
y='Avg_MPG_City').show()
```

## Python Code (Text+Emoji)

```
import 🐼 , 🦆 , 📊 .express as 📊

from 📅 ⌚ import 📅 ⌚
🖨️ ( 📅 ⌚ .now().strftime("%Y-%m-%d"))

🤔 📅 ⌚ .now().weekday() in (5, 6):
    🖨️ ("It's the weekend!")
🚫 :
    🖨️ ("It's a work day!")

📊 .bar( 🚗 _summary, x='Type',
y='Avg_MPG_City').show()
```



# SQL

duckdb.sql("select type, avg(MPG\_City) from cars group by 1 order by 2 desc")

 .sql("  type, avg(MPG\_City) from   $\Sigma$  1  2  ")

SELECT (DUCKDB)

## SQL Code (Text)

```
import pandas, duckdb
df_cars=pandas.read_csv('cars.csv')
cars_summary=duckdb.sql("""
select type,
avg(MPG_City) as avg_MPG_City,
avg(MSRP) as avg_MSRP
from df_cars
group by 1 order by 2 desc, 1
""").df()
print(cars_summary)
```

## SQL Code (Text+Emoji)

```
import 🐼 , 🦆
df_🚗 = 🐼 .read_csv('🚗.csv')
🚗_summary= 🦆 .sql("""
🛒 type,
avg(MPG_City) as avg_MPG_City,
avg(MSRP) as avg_MSRP
from df_🚗
Σ 1 2 ⬆️ 2 ⬇️ , 1
""").df()
🖨️ ( 🚗 _summary)
```

WHERE'S THE BEEF?,"



**“Cute, but are there  
any real benefits?”**



```
{ "Insert Stuff": {
    "prefix": "inline",
    "body": ["# Connect to Snowflake",
"conn = snowflake.connector.connect(",
"account='your_account'",
"user='your_username'",
"password='your_password')", ],
    "description": "Insert lines" }}
```

1  1.169

```
# Connect to Snowflake
conn = snowflake.connector.connect(
    account='your_account',
    user='your_username',
    password='your_password')
'''
```

Do you like to use keyboard shortcuts and code snippets? Emoji-mapped code can offer the same keystroke reduction benefit, **PLUS** allows code to be maintained at the single emoji level instead of at the fully-expanded code level – here that means 1 emoji vs. 5 lines of code to connect to a Snowflake database.

The emoji mapping in effect provides a similar kind of functionality as a simple, single character macro.

```
from xml.etree.cElementTree import ElementTree, ParseError
def parse_xml_from_file(filename):
    try:
        data = ElementTree().parse(filename)
    except ParseError:
        print('Failed to parse file {}'.format(filename))

filename = r'C:\Data.xml'
parse_xml_from_file(filename)
```

Module name

Class name

Function name

Parameter name

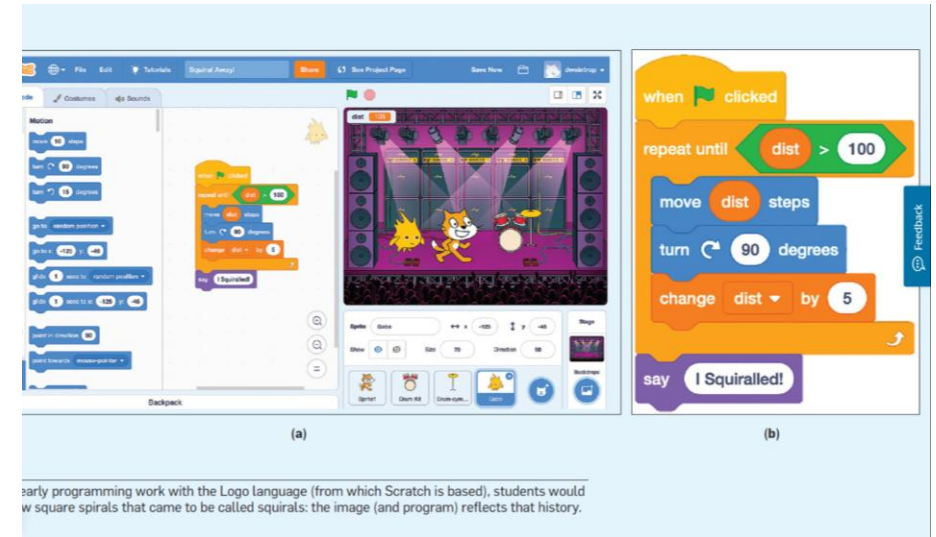
# Syntax Highlighting

Like using syntax highlighting and code-naming standards, e.g., CamelCase, Hungarian notation?

Emoji mapped to text could represent all or part of your package names, package aliases, variable names, function names, constants, etc. Same improved readability, **PLUS** emojis will help make names 'pop' even more!

# Make Coding More Playful & Approachable!

For those new to text programming (e.g., kids), letting them use emoji-to-text mappings – both pre-defined and of their own creation – might help get them to embrace coding, ease the transition from block to text coding, and perhaps even help some grasp algebra concepts before they're deemed "ready" for that subject!

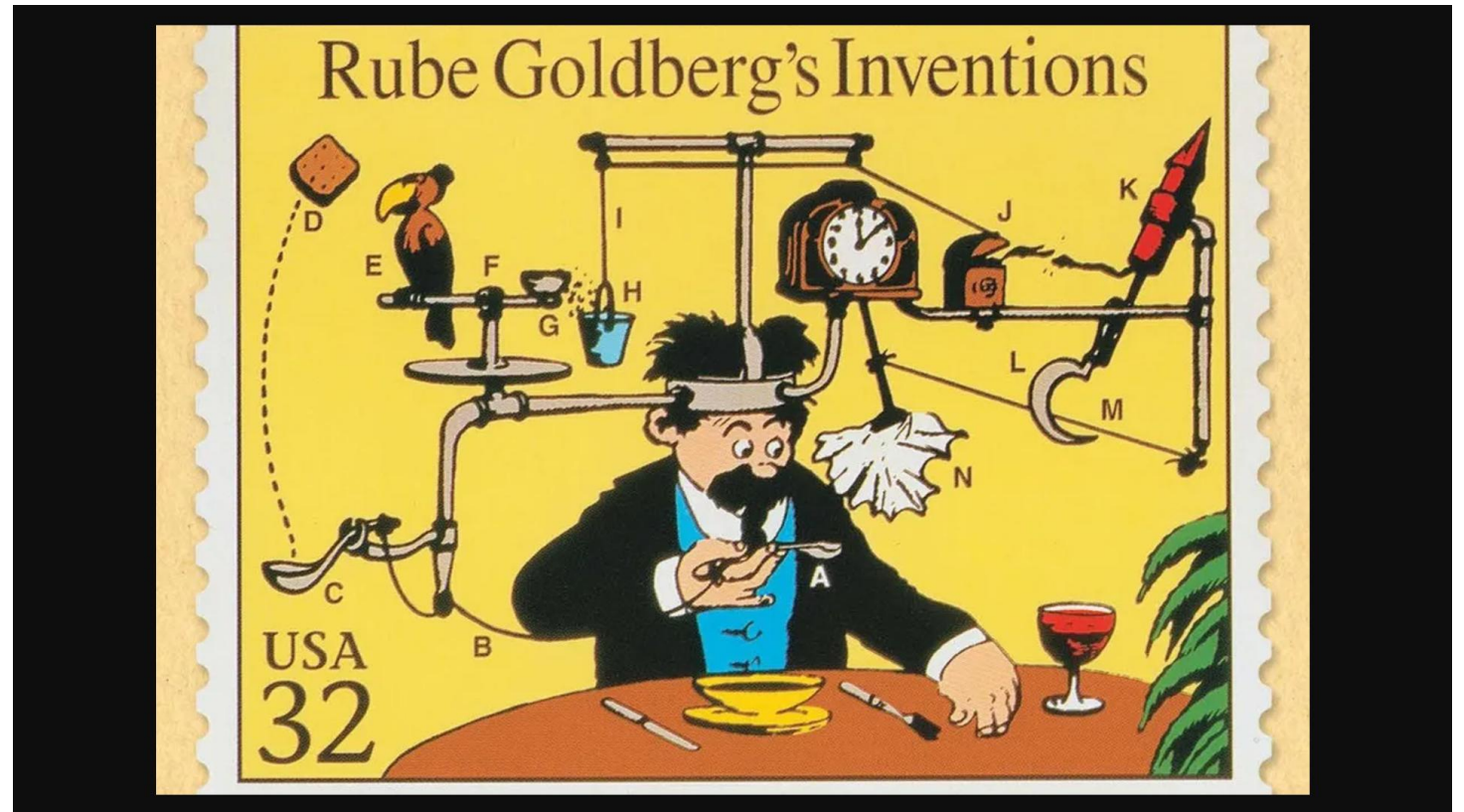


```
src > quizzes > tests > display_tests.py > DisplayQuizViewTests > test_quiz_redirects
4 from django.test import TestCase
5 from django.urls import reverse
6
7
8 class DisplayQuizViewTests(TestCase):
9     def test_quiz_404(self):
10         url = reverse("quizzes:display_quiz", args=(12,))
11         response = self.client.get(url)
12         self.assertEqual(response.status_code, 404)
13
14     def test_quiz_redirects(self):
15         quiz, question, _ = create_quiz()
16         url = reverse("quizzes:display_quiz", args=(quiz.pk,))
17         response = self.client.get(url)
18         self.assertRedirects(response, reverse("quizzes:display_question", args=(quiz.pk, question.pk)))
19
20
21 class DisplayQuestionViewTests(TestCase):
22     def test_quiz_404(self):
23         url = reverse("quizzes:display_question", args=(12, 1))
24         response = self.client.get(url)
25         self.assertEqual(response.status_code, 404)
26
27     def test_question_404(self):
28         quiz, question, _ = create_quiz()
29         url = reverse("quizzes:display_question", args=(quiz.pk, question.pk + 100))
30         response = self.client.get(url)
```

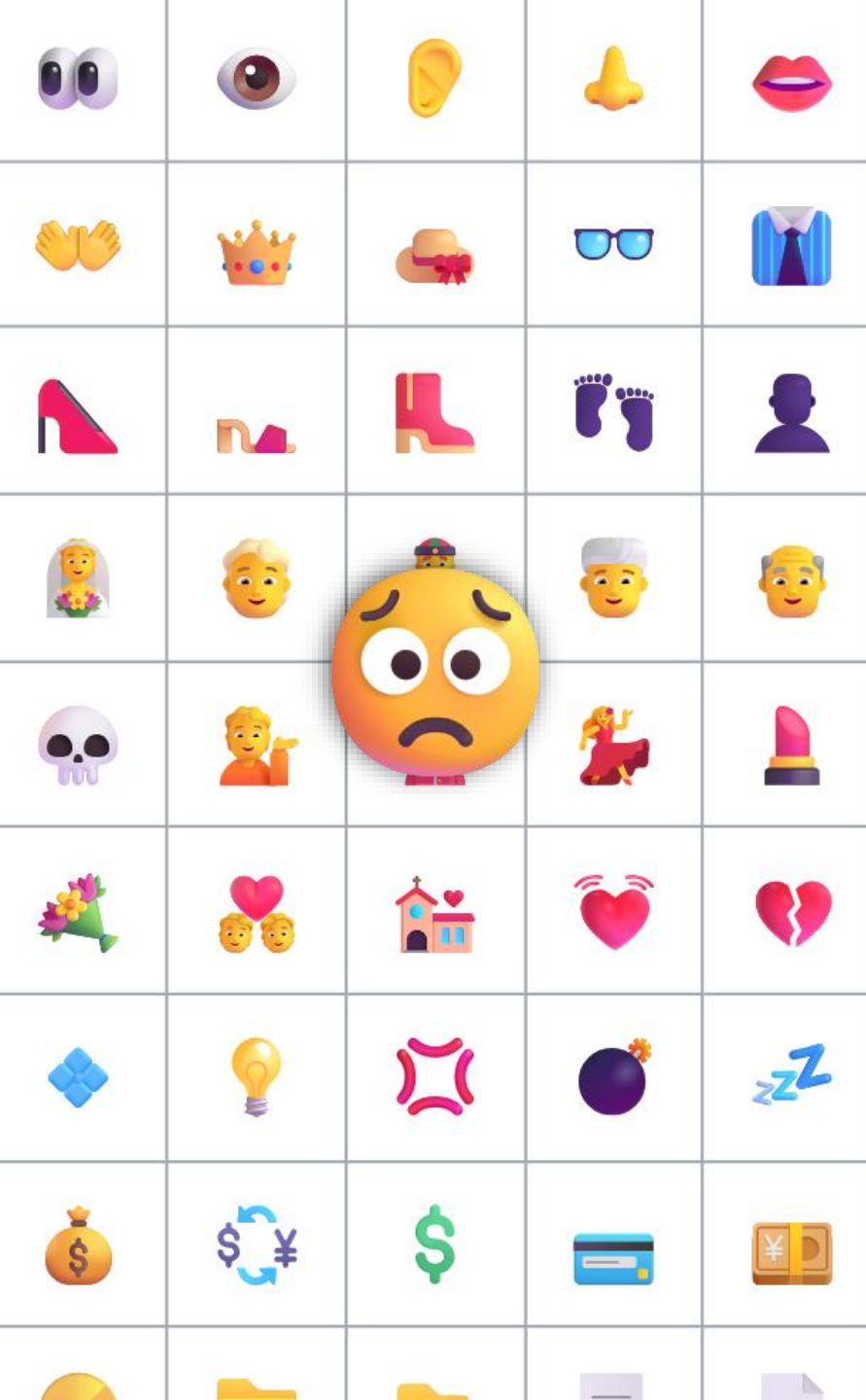


Proof-of-Concept

# A Simple Emoji-to-Text Preprocessor



- Translates emoji-laden code into all text for Python
- Uses an IPython input transformer with VS Code + Jupyter Extension
- Just a quick-and-dirty POC, ignores messiness of real-world language design and implementation!



# Emojis are complicated!



Man Running Facing Right

| Code Points (Sequence)                                                                          | UTF-8 Encoding (Hex) |
|-------------------------------------------------------------------------------------------------|----------------------|
| U+1F3C3 [  ] |                      |
| U+200D                                                                                          |                      |
| U+2642 [ ♂ ]                                                                                    | F0 9F 8F 83 E2 80    |
| U+FE0F                                                                                          | 8D E2 99 82 EF B8    |
| U+200D                                                                                          | 8F E2 80 8D E2 9E    |
| U+27A1 [ ➡ ]                                                                                    | A1 EF B8 8F          |
| U+FE0F                                                                                          |                      |

# Emoji Encoding

- In the Unicode and UTF-8 world of emoji, **one character may now occupy 1-4 bytes**
- Complex graphemes built from **multiple code points, joiners & modifiers** require even more!

| Character | Description                        | Code Points (Sequence)                                                          | UTF-8 Encoding (Hex)                                                          | Bytes |
|-----------|------------------------------------|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------|-------|
| A         | Latin Capital A                    | U+0041                                                                          | 41                                                                            | 1     |
| 🤔         | Thinking Face                      | U+1F914                                                                         | F0 9F A4 94                                                                   | 4     |
| ✅         | White Heavy Check Mark             | U+2705                                                                          | E2 9C 85                                                                      | 3     |
| ❌         | Cross Mark Button                  | U+274E                                                                          | E2 9D 8E                                                                      | 3     |
| 🏃         | Person Running                     | U+1F3C3                                                                         | F0 9F 8F 83                                                                   | 4     |
| 🏃         | Man Running Facing Right (ZWJ seq) | U+1F3C3 [🏃]<br>U+200D<br>U+2642 [♂]<br>U+FE0F<br>U+200D<br>U+27A1 [➡]<br>U+FE0F | F0 9F 8F 83 E2<br>80 8D E2 99 82<br>EF B8 8F E2 80<br>8D E2 9E A1 EF<br>B8 8F | 22    |



# Houston, We Have an Emoji Problem



## Bad News:

Python has native emoji support but

- 😱 Limited to single codepoint emoji
- 😱 No support for zero-width joiners
- 😱 No support for variation selectors



## Good News:

Python packages fill in the gaps

- 😊 Easily identify and extract emojis
- 😊 Supports complex emoji ‘clusters’
- 😊 Use [regex](#) package with /X options

```
import regex, pandas as pd
dict = {'🤔': 'if', '✅': 'then', '❌': 'else', '🏃': 'run;'}
str="🤔 a=b ✅ c=d; ❌ e=f; 🏃 "
df = pd.DataFrame([match.group() for match in regex.finditer(r"\X", str)],
                  columns=['grapheme'])
df['text'] = df['grapheme'].map(dict)
df['text'].fillna(df['grapheme'], inplace=True)
df
```

| # | Grapheme | Text | #  | Grapheme | Text | #  | Grapheme | Text | #  | Grapheme | Text |
|---|----------|------|----|----------|------|----|----------|------|----|----------|------|
| 0 | 🤔        | if   | 6  | ✅        | then | 12 |          |      | 18 | ;        | ;    |
| 1 |          |      | 7  |          |      | 13 | ❌        | else | 19 |          |      |
| 2 | a        | a    | 8  | c        | c    | 14 |          |      | 20 | 🏃        | run; |
| 3 | =        | =    | 9  | =        | =    | 15 | e        | e    |    |          |      |
| 4 | b        | b    | 10 | d        | d    | 16 | =        | =    |    |          |      |
| 5 |          |      | 11 | ;        | ;    | 17 | f        | f    |    |          |      |

# A Python Emoji Preprocessor



Python code with emojis



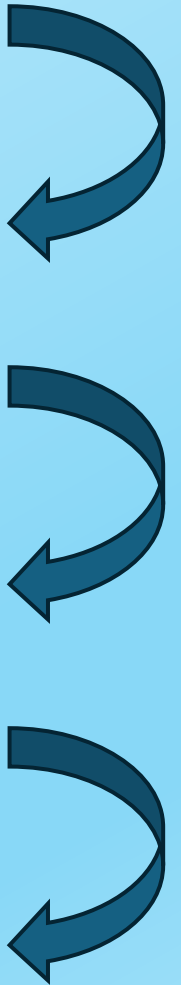
IPython transformer replaces emojis with text (via regex)



Python code without emojis

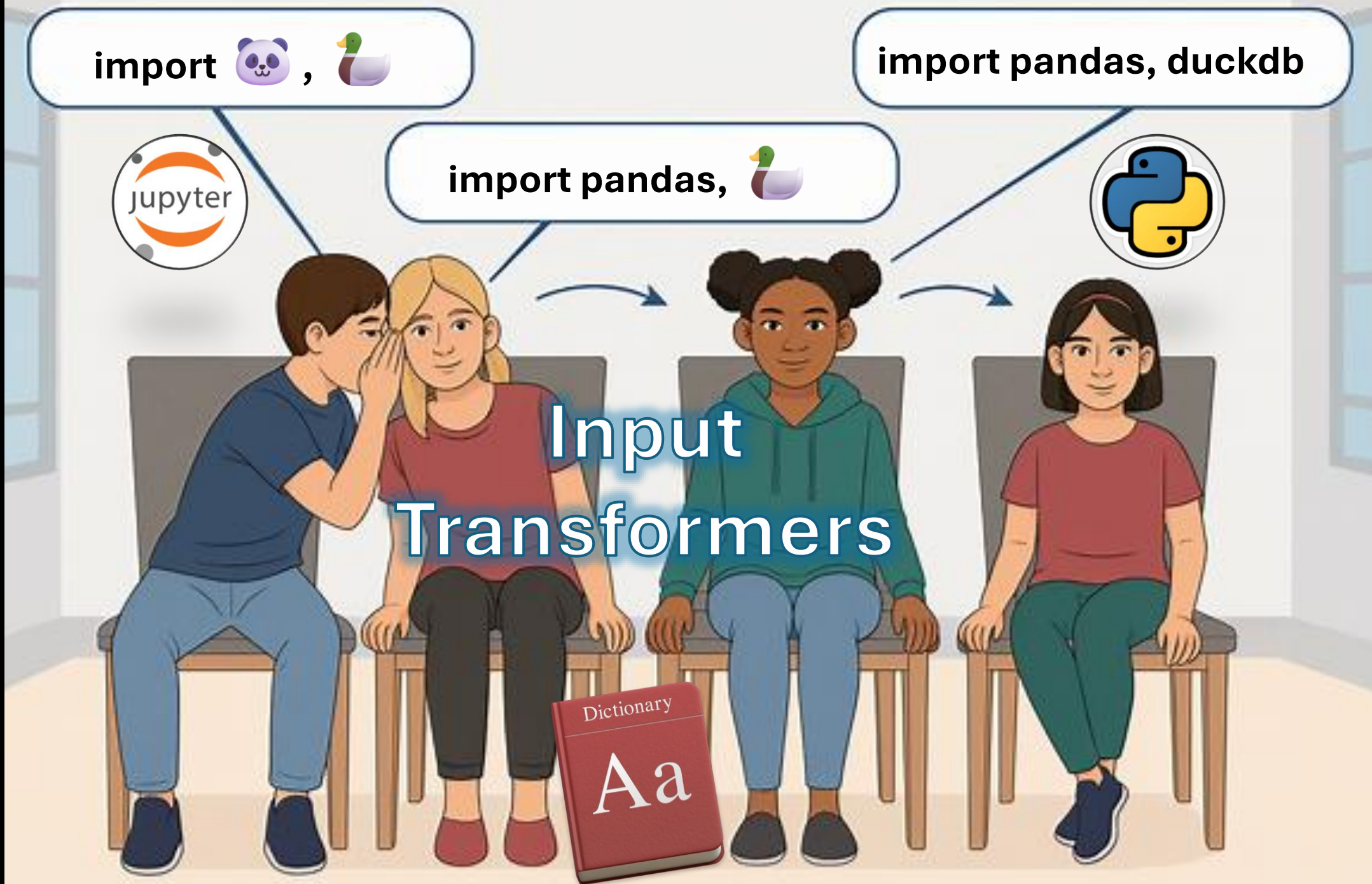


Python runs emoji-free code





# IPython Custom Input Transformation



```
dict = {'🤔': 'if', '❌': 'else', '💾': 'data',  
        '🐼': 'pandas', '🦆': 'duckdb',  
        '📊': 'plotly', '💾': 'data',  
        '📅': 'date', '🕒': 'time', '🖨️': 'print',  
        '🗄️': 'create table', '🗑️': 'drop table',  
        '🛒': 'select', '⬅️': 'from',  
        '⬆️': 'asc', '⬇️': 'desc',  
        '🔗': 'join', '↕️': 'order by',  
        'Σ': 'group by'}
```

**# Define emoji-to-text mapping, set flag indicating translation is active**

```
global dict, emoji2txt
globals()['emoji2txt']=False
```

```
dict = {
'😬': 'if', '❌': 'else', '🖨️': 'print', '🐼': 'pandas', '🦆': 'duckdb', '📊': 'plotly',
'🔤': 'str', '💾': 'data', '📅': 'date', '🕒': 'time', '🔄': 'while',
'🗄️': 'create table', '🗑️': 'drop table', '🛒': 'select', '👈': 'from', '🔗': 'join',
'⬆️': 'order by', '⬇️': 'asc', '⬇️': 'desc', 'Σ': 'group by',
'🚗': 'cars',
'🔌': '"""# Connect to Snowflake
conn = snowflake.connector.connect(
account='your_account',
user='your_username',
password='your_password')
"""}

```

```
globals()['emoji2txt']=True
```



```
import regex, json, pandas as pd
```

### **# Parse submitted Python code for emojis, other characters**

```
def grapheme_positions(text):  
    matches = regex.finditer(r'\X', text)  
    return [{'grapheme': m.group()} for m in matches]
```

### **# Replace emojis in Python code with associated text from global dictionary**

```
def emoji2text(lines):  
    global emoji2txt  
    global dict  
    if globals()['emoji2txt'] is None:  
        emoji2txt=False  
    if emoji2txt==False:  
        return lines  
    df = pd.DataFrame.from_dict(grapheme_positions(''.join(lines)), orient='columns')  
    df['replstr'] = df['grapheme'].map(dict)  
    df['replstr'].fillna(df['grapheme'], inplace=True)  
    return ''.join(df["replstr"].tolist()).splitlines(keepends=True)
```

### **# Add input transformer**

```
ip = get_ipython()  
ip.input_transformers_cleanup.append(emoji2text)
```

```
import pandas, duckdb, plotly.express as plotly
from datetime import datetime
```

PACKAGES

```
print(datetime.now().strftime("%Y-%m-%d"))
if datetime.now().weekday() in (5, 6):
    print ("It's the weekend!")
else:
    print("\nIt's a work day!")
```

PYTHON

```
df_cars=pandas.read_csv('cars.csv')

cars_summary=duckdb.sql("""
select type, avg(MPG_City) as Avg_MPG_City, Avg(MSRP) as avg_MSRP
from df_cars group by 1 order by 2 desc, 1
""")
print("\n",cars_summary)
```

SQL

```
plotly.bar(cars_summary, x='Type', y='Avg_MPG_City').show()
```

PLOTLY

```
import 🐼, 🦆, 📈.express as 📈
from 📅 ⌚ import 📅 ⌚
```

PACKAGES

```
🖨️ ( 📅 ⌚ .now().strftime("%Y-%m-%d"))
🤔 📅 ⌚ .now().weekday() in (5, 6):
    🖨️ ("It's the weekend!")
🟢 ✕ :
    🖨️ ("\nIt's a work day!")
```

PYTHON

```
df_ 🚗 = 🐼 .read_csv(' 🚗 .csv')
```

```
🚗 _summary= 🦆 .sql("""
🛒 type, avg(MPG_City) as Avg_MPG_City, Avg(MSRP) as avg_MSRP
from df_ 🚗 Σ 1 ⬆ 2 ⬇ , 1
""
).df()
🖨️ ("\n", 🚗 _summary)
```

SQL

```
📈 .bar( 🚗 _summary, x='Type', y='Avg_MPG_City').show()
```

PLOTLY



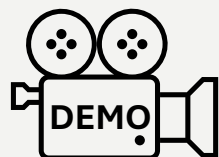
## EXPLORER

✓ NO FOLDER OPENED

You have not yet opened a folder.

Open Folder

Opening a folder will close all currently open editors. To keep them open, [add a folder](#) instead.



## > OUTLINE

## > TIMELINE

demo4pydata.ipynb ✕

C: > Users > tedco > OneDrive > Pydata2025 >  demo4pydata.ipynb >  # Remove existing input transformer

 Generate
  Code
  Markdown
  Run All
  Restart
  Clear All Outputs
  ...

 base (Python 3.10.9)

```
# Remove existing input transformer
ip.input_transformers_cleanup.remove(emoji2text)
```

## Python

```
# Create global emoji-to-text Python dictionary, set global indicator that translation is active
global dict, emoji2txt
globals()['emoji2txt']=False
dict = {
'😞':'if', '❌':'else', '🖨':'print',
'🐼':'pandas', '🦆':'duckdb', '📈':'plotly', '📊':'excel',
'🔤':'str', '💾':'data', '📅':'date', '🕒':'time', '🔄':'while',
'🗄':'create table', '🗑':'drop table', '🛒':'select', '⬅':'from',
'🔗':'join', '📶':'order by', '⬇':'asc', '⬇':'desc', 'Σ':'group by',
'🚗':'cars'}
globals()['emoji2txt']=True
```

## Python

```
import regex, json, pandas as pd
```



## A GPT-5 Second Opinion

*“Give me some pros and cons of using emojis to represent certain statement names, variable names, function names, literals, and code snippets in Python...”*

*“Create a summary table explaining if each proposed use is Pythonic or not Pythonic...”*



## GPT-5 Generated “Verdict Table”

| Use Case                                                | Possible?            | Pythonic? | Notes                                                                               |
|---------------------------------------------------------|----------------------|-----------|-------------------------------------------------------------------------------------|
| String Literals (" 🌍 ")                                 | ✅ Yes                | ✅ Yes     | Safe and widely used                                                                |
| Comments with Emojis                                    | ✅ Yes                | ⚠️ Okay   | Decorative only; use sparingly                                                      |
| Variable Names ( 🔥 )                                    | ✅ Yes                | ❌ No      | Works, but unreadable, flagged by linters                                           |
| Function Names ( 🐍 )                                    | ✅ Yes                | ❌ No      | Technically valid, but violates clarity                                             |
| Emojis via IPython Transformer                          | ✅ Yes                | ❌ No      | Hacky DSL layer; non-standard                                                       |
| Package Names ( 🔥 .py)                                  | ✅ Yes                | ❌ No      | Technically importable, but breaks portability and readability                      |
| Emoji Macros ( 🔌 → code snippet)                        | ✅ Yes                | ❌ No      | Fun for demos; expands to full code, but non-standard and fragile                   |
| Complex Grapheme ( 🏃 )                                  | ⚠️ Sometimes         | ❌ No      | Can expand via transformer if exact ZWJ + selectors match; fragile across editors   |
| Emoji in place of Python keywords via input transformer | ✅ Yes (IPython only) | ❌ No      | Possible by preprocessing input; playful DSL trick, but non-standard & not portable |



## GPT-5 Generated “Verdict Table”

| Use Case                                                | Possible?           | Pythonic? | Notes                                                                               |
|---------------------------------------------------------|---------------------|-----------|-------------------------------------------------------------------------------------|
| String Literals ( " 🌍 ")                                | ✅ Yes               | ✅ Yes     | Safe and widely used                                                                |
| Comments with Emojis                                    | ✅ Yes               | 👉 Okay    | Decorative only, use sparingly                                                      |
| Variable Names ( 🔥 )                                    | ✅ Yes               | ❌ No      | Works but unreadable, flagged by linters                                            |
| Function Names ( 🐍 )                                    | ✅ Yes               | ❌ No      | Technically valid, but violates clarity                                             |
| Emojis via IPython Transformer                          | ✅ Yes               | ❌ No      | Tricky DSL layer; non-standard                                                      |
| Package Names ( 🍌 .py)                                  | ✅ Yes               | ❌ No      | Technically importable, but breaks portability and readability                      |
| Emoji Macros ( 🔌 → code snippet)                        | ✅ Yes               | ❌ No      | Fun for demos; expands to full code, but non-standard and fragile                   |
| Complex Grapheme ( 🏃 )                                  | ⚠️ Sometimes        | ❌ No      | Can expand via transformer if exact ZWJ + selectors match; fragile across editors   |
| Emoji in place of Python keywords via input transformer | ✅ Yes (Python only) | ❌ No      | Possible by preprocessing input; playful DSL trick, but non-standard & not portable |



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By [Jennifer Goforth Gregory](#)  
Posted Dec 5 2025



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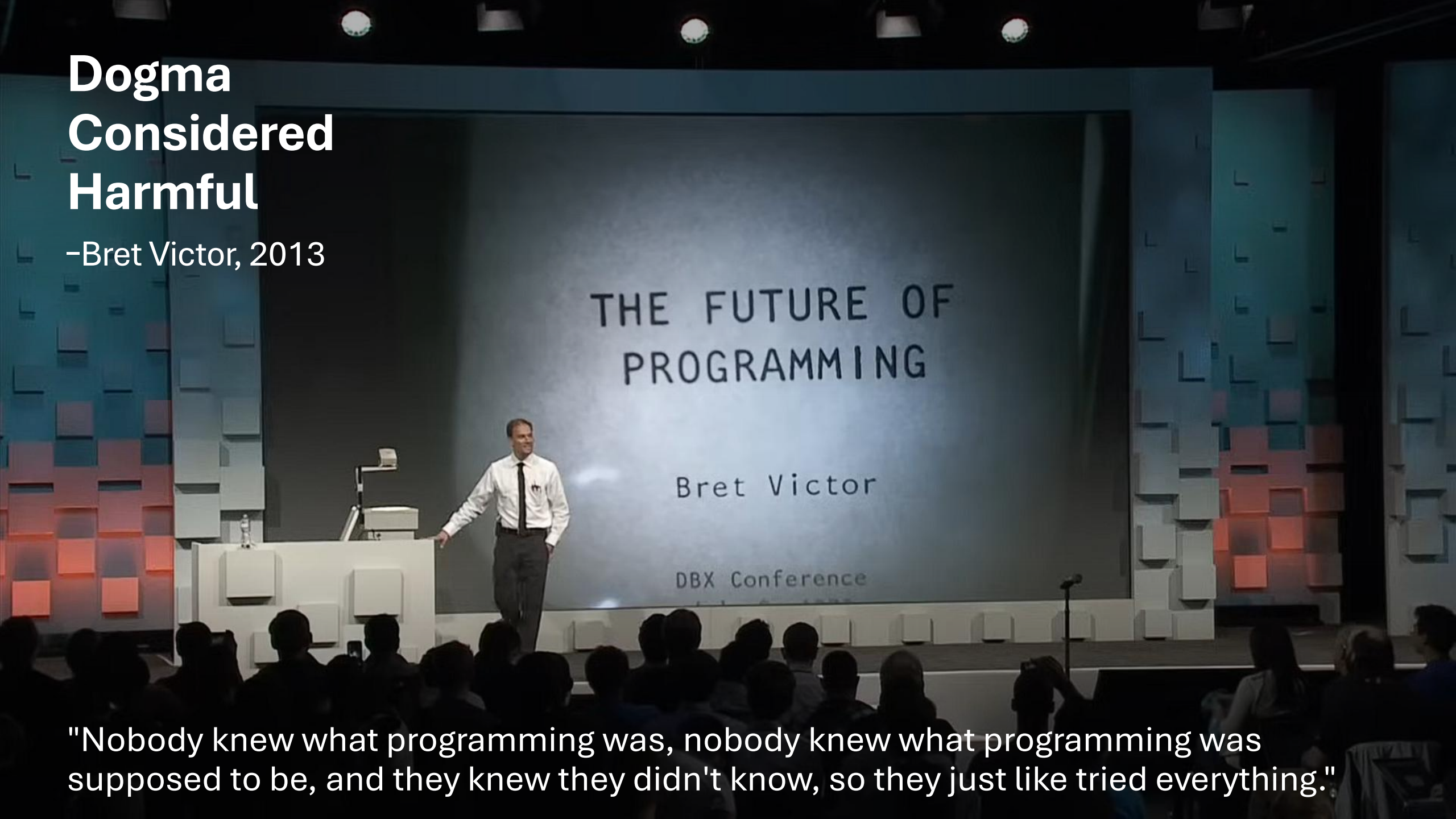
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# Dogma Considered Harmful

-Bret Victor, 2013



## THE FUTURE OF PROGRAMMING

Bret Victor

DBX Conference

"Nobody knew what programming was, nobody knew what programming was supposed to be, and they knew they didn't know, so they just like tried everything."

Don't be  
'That Guy'

## BINARY

```
1999015201 5801610167  
0170017301 7601820185  
0141014401 5001530156  
0154020402 0501630254  
0211014201 4901720143  
0203620602 1901770235  
0153015502 7201930253
```

Programming

## FORTRAN

```
C 0000 RECTANGULAR MATRIX  
C 0000 MULTIPLICATION  
DIMENSION A(4*5), B(5*3)  
READ 1, A+B  
READ 1, N+M/L  
7 00 4 1 1 N
```

NOT  
Programming

```
0000000000 0000000000  
6501640167 6901701822  
6901851822 01680141  
6901961823 00010162  
6800010169 40070211  
1500390143 0040203  
6580060231 90192  
3500040221 58002  
3901740221 00137  
2401100221 0201  
691661822 0272  
01871822 0010160  
4601460221 0010162  
460174017 0008000  
00000000 00000000  
00000001 00000269  
00000000 00001999
```



I think you have to say,  
“We don’t know what  
programming is...”





# QUESTIONS?

Ted Conway

Chicago, IL

ted.j.conway@gmail.com

@vivasasvegas X 🦋



Slides available at:

[github.com/tedconway/pydataboston2025](https://github.com/tedconway/pydataboston2025)

